TeamSPRC

SPRC Wizard

How to install it and set up a combat robot, for RadioMaster MT12, Pocket, TX15, TX16S and TX16S MK3 radios on EdgeTX

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Before you start

SPRC Wizard is a tool on the radio that builds a complete combat robot model in one go. The left wheel goes on CH1, the right wheel on CH2 and the weapon on CH3, with rate and expo at the team's standard values and a drive mode for when the robot is upside down.

You need:

- A RadioMaster **MT12** or **Pocket** on EdgeTX 2.10, or a **TX15**, **TX16S** or **TX16S MK3** (read the TX15 and TX16S section first)
- A Windows PC and the radio's USB cable, or an SD card reader
- This zip file, extracted first: right-click it and choose Extract All

Take the blade or weapon belt off every time you set up or test the robot, and put it on a stand with the wheels off the ground. If the weapon signal polarity is set wrong, the weapon spins the moment you plug in the battery.

Install to the card from a Windows PC

1 Connect the radio's card to the PC

Turn the radio on and plug the USB cable into the PC. The radio asks how to use USB: choose **USB Storage (SD)**. You can also take the SD card out and use a card reader. The card appears as a new drive in File Explorer.

2 Open SPRC_Install.exe

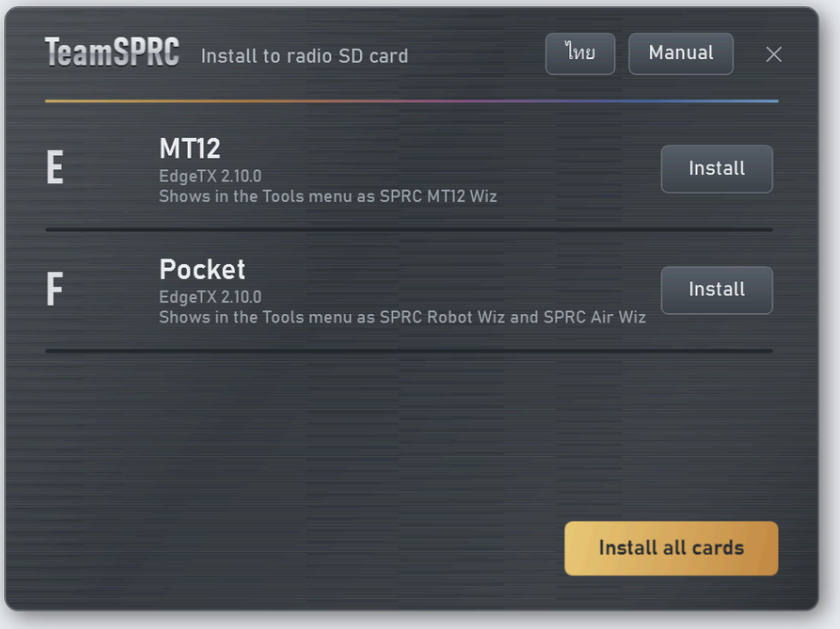
Double-click the file with the steel plate SPRC icon, the same one at the top of this manual. The program starts in Thai. Press **English** at the top right to switch the program and this manual to English. It remembers the choice. The **Manual** button beside it opens this manual at any time, with no internet needed.

When it opens, the program checks for updates by itself. With internet you get the latest wizard and manual without downloading the program again. If a newer program is out, the window closes and reopens by itself once. The install buttons work once the check finishes, which takes a few seconds. Without internet you can still install the version inside the program.

> The first time, some PCs show a blue "Windows protected your PC" box because this program has no paid digital signature. Press **More info**, then **Run anyway**.

3 Wait for the card to appear in the window

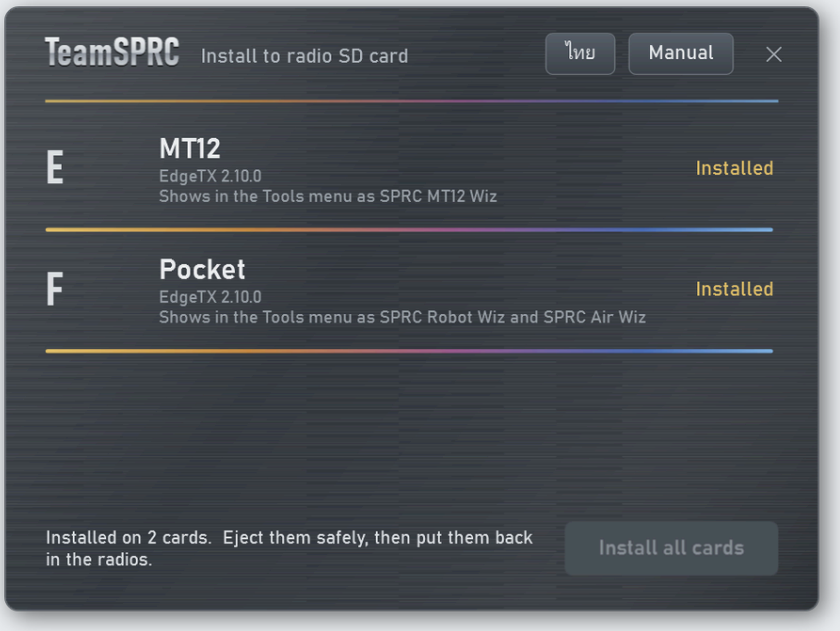
Each row is one card and shows the radio model, the EdgeTX version and the menu name you will see on the radio. If you plug the card in after opening the program, the row appears by itself within seconds.



MT12 and Pocket cards found together. Each gets the wizard for its own radio.

4 Press Install

Press **Install** on the card's row, or **Install all cards** if several are plugged in. When it finishes, the row says Installed and the line under it turns the rainbow colours of a weld.



Installed. The bottom bar tells you to eject safely before putting the card back in the radio.

The installer writes one wizard file to the `SCRIPTS/TOOLS` folder. It leaves your models, ExpressLRS and EdgeTX's Wizard Loader alone.

5 Eject before you unplug

In File Explorer, right-click the card's drive and choose Eject. Then unplug the USB cable, or take the card out and put it back in the radio. If you pull it out without ejecting, the file may not be fully written.

Not on Windows? Copy the file yourself

The `manual` folder in the zip has both wizards. Copy the one for your radio to `SCRIPTS/TOOLS/` on the card.

Radio	File	Name in the Tools menu
MT12	`mt12_wizard.lua`	SPRC MT12 Wiz
Pocket, TX15, TX16S, TX16S MK3	`pocket_wizard.lua`	SPRC Robot Wiz
Also on Pocket, TX15, TX16S, TX16S MK3	`air_wizard.lua`	SPRC Air Wiz

If that folder has a file with the same name ending in `.luac`, delete it, or the radio will run the old version.

Open the wizard on the radio

The wizard deletes all INPUTS and MIXES of the selected model. Use it only on a new, empty model. Never run it over a model you have already set up.

1 Create a new model

Press `MDL`, pick an empty model slot and create a new model. Name it after the robot now so you don't mix them up.

2 Open it from the Tools menu

Press `SYS`. The first page is TOOLS. Choose `SPRC MT12 Wiz` on MT12 or `SPRC Robot Wiz` on the others, then press `ENTER`. On newer EdgeTX, TOOLS may be called Apps.

3 On MT12, choose the type first



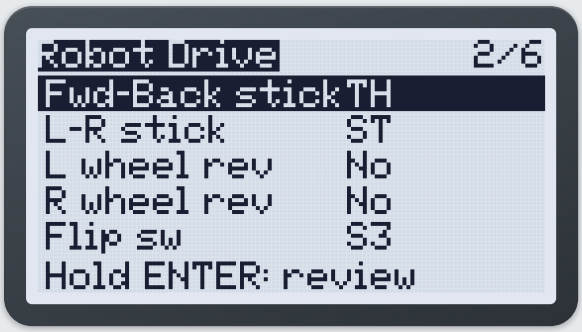
Simulated screen from the same wizard that is in the zip. The text matches the real screen; the lettering may look slightly different.

MT12 has five types: Car, Crawler, Tank, Boat and Robot. Scroll to `Robot` and press `ENTER`.

The other radios have no such page. They go straight to the robot settings.

This manual covers Robot. The other types use the same buttons.

Settings



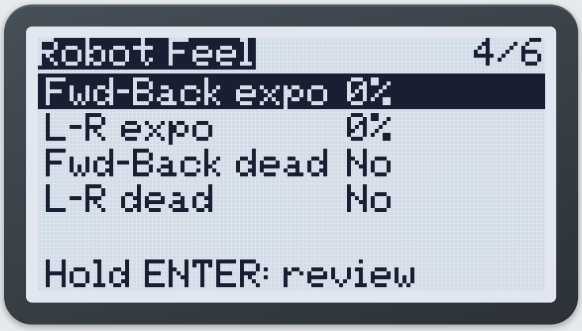
The Drive page of Robot on MT12

The options are on four pages: Drive, Rates, Feel and Weapon. The selected row is a full black bar. Scroll past the last row and the next page appears by itself. The numbers at the top right show which page you are on.

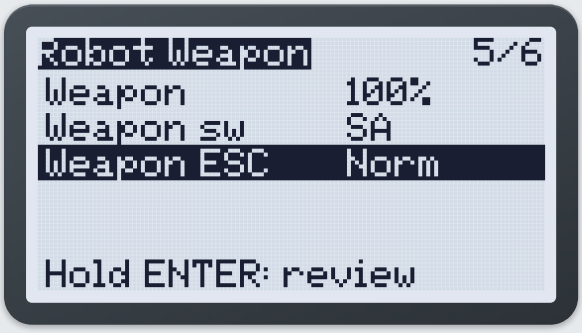
- Scroll wheel: move up and down
- `ENTER`: start editing. The value blinks. Scroll to change it, then press `ENTER` again to finish.
- `RTN`: cancel the edit, or leave the wizard
- Hold `ENTER`: go to the channel review



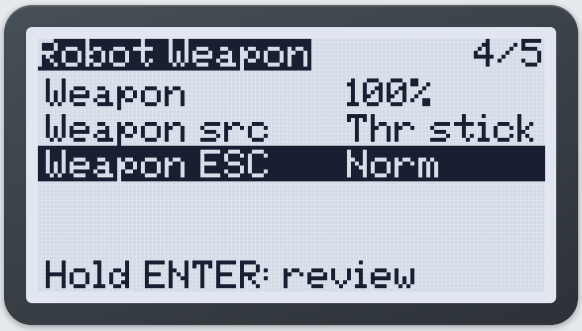
Editing Fwd-Back rate. The black bar goes away and the value blinks until you press ENTER.



The Feel page sets expo and Dead zone separately for each stick.



The last page is the weapon.



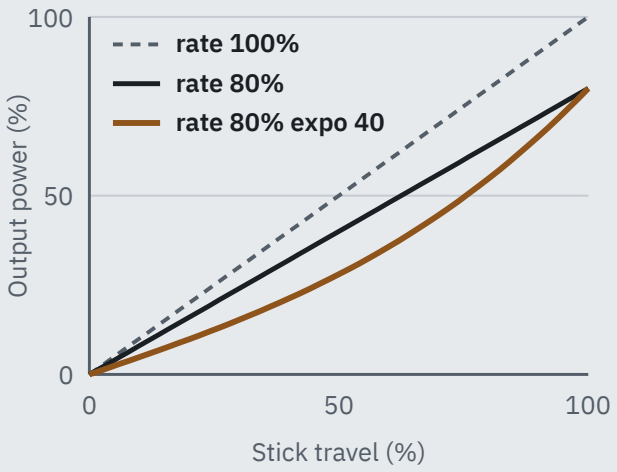
On Pocket the weapon starts on the Thr stick, and you drive with the right stick (Ele and Ail).

What each option does

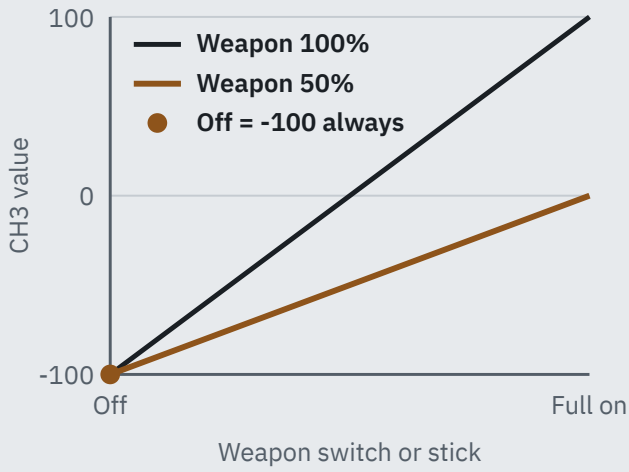
The table shows the default values the wizard sets, which are the team's standard. Most need no change.

Option	MT12	Pocket	What it does
Drive	none	Arcade	How you drive. Arcade: one stick does both forward and steering. Tank: two sticks, one per wheel.
Fwd-Back stick	TH	Ele	The stick for forward and backward. In Tank this row becomes L wheel stick.
L-R stick	ST	Ail	The stick for left and right. In Tank this row becomes R wheel stick.
L wheel rev	No	No	Reverses the left wheel. Use it when that motor runs backward, for example when its wires are soldered the other way round.
R wheel rev	No	No	Reverses the right wheel.
Flip sw	S3	SA	The switch for upside-down mode. Up is normal driving, down is flipped. No means unused.
Fwd-Back rate	80%	80%	Top power for forward and backward. Lower it if the robot is too fast to control.
L-R rate	50%	50%	Steering sensitivity. Lower it if the robot spins too fast.
Turn rate	100%	100%	How much steering goes into the wheel mix. Usually leave it at 100% and adjust L-R rate

Option	MT12	Pocket	What it does
			instead.
Fwd-Back expo	0%	0%	Softens the middle of the stick. The end of the stick still reaches the full rate. 0 means a direct response.
L-R expo	0%	0%	The same, for steering.
Fwd-Back dead	No	No	Dead zone of the forward-back stick: the range around center that has no effect. No means unused.
L-R dead	No	No	Dead zone of the steering stick, set separately from the forward-back stick.
Weapon	100%	100%	Top weapon power. Lowering the % leaves the off position fully off. No means no weapon channel.
Weapon sw or **Weapon src**	SA	Thr stick	The switch that turns the weapon on. On Pocket you can also pick the Thr stick, which lets you bring the weapon up gradually.
Weapon ESC	Norm	Norm	Signal polarity of the weapon ESC, which has nothing to do with spin direction. Choose Rev if your ESC treats the other end as off.



Rate lowers the maximum along the whole line. **Expo** makes the first part of the stick rise slowly for finer control, while the end of the stick still reaches the rate.



Weapon % lowers only the ceiling. The off position stays at -100, so the weapon stops completely when the switch is off.

Upside-down mode (Flip sw)

When the robot is flipped, the wheels touch the ground on the opposite side, so pushing the stick forward makes it drive backward. The Flip switch fixes this: up is normal driving, down is flipped. The default is S3 on MT12, SA on Pocket and SF on TX16S. Change it on the **Flip sw** row.

Flip switch	Push the stick forward	Move the stick left
Up (normal)	The robot drives forward	The robot turns left
Down (flipped)	The robot still drives forward as the driver sees it	The robot still turns left, with no need to think in reverse

The wizard reverses forward and backward only. Steering needs no reversal, because the left and right wheels have already swapped sides when the robot is flipped.

This row offers no 3-position switches, because the middle position would cut all forward power. SB, SC and SD on MT12 are spring-return buttons. You can choose them, but Flip then works only while you hold the button.

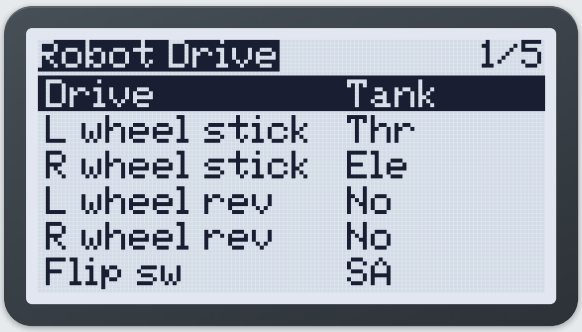
Dead zone

Dead zone is a range around the stick's center that has no effect. You set it separately on the Feel page: **Fwd-Back dead** for the forward-back stick and **L-R dead** for the steering stick. The value is a percentage of stick travel, up to 25%. Past that range the power rises gradually up to the rate, and expo works as before. Start small: the bigger the value, the farther you must push the stick before the robot moves.

Use it when the stick doesn't return exactly to center and the robot creeps. If the wheels creep while the channel reads 0, the ESC's center point is off. Fix that with Subtrim or by calibrating the ESC, because Dead zone can't fix it.

This option writes to curves CV1 and CV2 of the model. Any curve you made yourself in those two slots gets overwritten.

Two-stick drive on Pocket (Tank)



Set the **Drive** row to Tank. The left stick controls the left wheel and the right stick the right wheel. The wizard assigns the Thr and Ele sticks for you and moves the weapon to switch SB, because the Thr stick is now used for driving.

Rows that have no effect in this mode disappear. That leaves Wheel rate and Wheel expo, which apply to both wheels, and a separate Dead zone for each wheel. Flip mode works as before: the left stick still controls the left side as the driver sees it.

The Thr stick must spring back to center. If it doesn't, the left wheel won't stop when you let go, and the robot may drive off as soon as it's powered on.

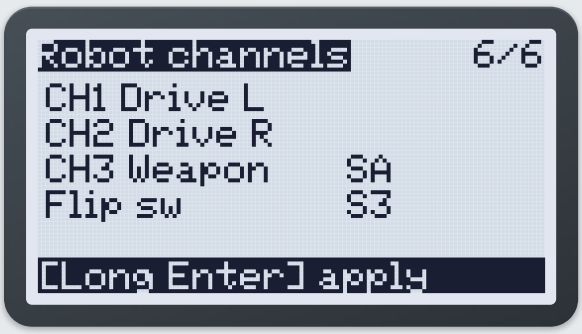
TX15 and TX16S

TX15, TX16S and TX16S MK3 have not been tried on a real radio yet. They have only been tested on a PC model. Try an empty model, then do every item in the checks before a match before you fit the weapon.

These three use the same wizard as Pocket, with the same options and defaults. The text scales to the screen. You still use the wheel or buttons and **ENTER**. The touch screen does nothing here yet.

The **Flip sw** row offers no switch set to 3 positions. On EdgeTX 2.12 and later the wizard reads the switch types from the radio's settings, and older firmware uses the factory layout. If the row starts at No, every switch is set to 3 positions, which can happen on a TX15. Set the switch you want to 2 positions on the radio's Hardware page, then open the wizard again. You can pick a front button for Flip, but the wizard won't choose one for you.

Review the channels and write to the model



Hold **ENTER** on a settings page to see what each channel does. **[Long Enter] apply** on the bottom bar means it's ready.

Hold **ENTER** again to write to the model, or press **RTN** to go back and edit.



If the bottom bar shows `SW CLASH: SB`, two functions are set on the same switch. In the picture, Flip sw and Weapon sw are both on SB, so the weapon would come on each time you throw the Flip switch.

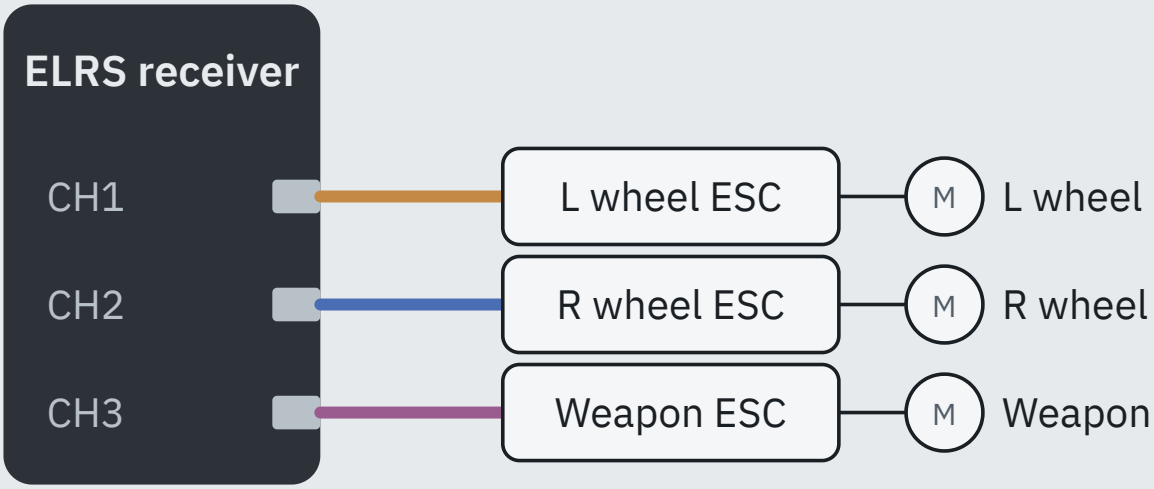
Press `RTN` and put them on different switches first. On Pocket you may see `CLASH: Thr stick` if the Thr stick is used for both driving and the weapon.



When it finishes, this screen reminds you that failsafe still has to be set on the receiver. Press `ENTER` or `RTN` to exit.

To change values later, run the wizard again on the same model, but anything you edited by hand in INPUTS or MIXES will be lost.

Wire up the receiver



On the wizard's review screen these three channels are called Drive L, Drive R and Weapon. Wire colours here only tell the channels apart. For the real wires, follow the polarity printed on the receiver.

Checks before a match

Put the robot on a stand with the wheels off the ground and remove the weapon before every item.

1 Wheel direction

Push the stick forward: both wheels must turn in the forward direction. Then move it left: the robot must turn left.

What you see	Fix
Forward makes one wheel turn backward	Set **L wheel rev** or **R wheel rev** for that side to Yes
Forward makes both wheels turn backward	Set both to Yes
Forward is right, but moving left turns the robot right	Swap the CH1 and CH2 plugs at the receiver

2 Try upside-down mode

With the Flip switch up, the robot must drive normally. Throw it down and push forward: the wheels must turn the opposite way from before. When you move the stick left, the wheels must turn the same as before you threw the switch. You can also watch the radio's Channels Monitor: when you push forward, CH1 and CH2 change sign, and when you steer they stay the same.

3 Weapon signal polarity

Turn the weapon switch off and plug in the robot's battery with the weapon still removed. The weapon motor must stay still. In Channels Monitor, CH3 must read -100 when off.

If the weapon motor spins by itself with the switch off, unplug the battery at once and set **Weapon ESC** to Rev.

4 Make your own Arm switch

The wizard doesn't create a weapon Arm switch. Add one yourself on the MIXES or Special Functions page, the way your team does it.

5 Set failsafe on the receiver

With ExpressLRS the radio doesn't send failsafe values, so you must set them on the receiver, through the ELRS Lua or the receiver's web page at 10.0.0.1. Set it so the wheels stop and the weapon shuts off. Then test by turning off the radio while the robot is on: the wheels must stop and the weapon must not spin.

6 Wheels creep when you let go of the stick

Adjust that channel's Subtrim on the OUTPUTS page until the wheel is still. If it still creeps only some of the time after you let go, raise that stick's Dead zone.

Planes, drones and helicopters

SPRC Air Wiz has not been tried on a real radio yet. Take the props or rotor blades off every time you test, and watch every channel in Channels Monitor before you put them back.



On Pocket, TX15, TX16S and TX16S MK3 the installer adds a second tool. Open **SPRC Air Wiz** from the Tools menu and choose Drone, Plane, Delta or Heli. The buttons work as in the robot wizard.

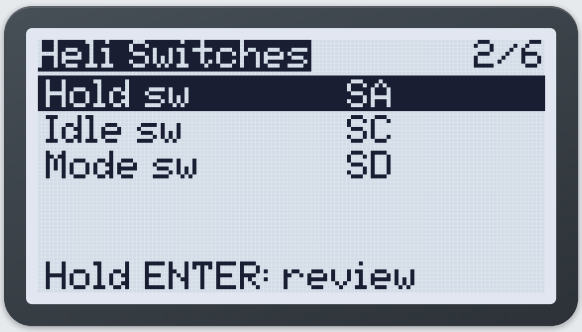
MT12 doesn't get this tool, because it has no Ail, Ele or Rud sticks.

Use it only on a new, empty model. The wizard deletes all INPUTS and MIXES, and Heli also overwrites curves CV1 to CV6.

Type	Channels	Options
Drone	CH1 to CH4 are Ail, Ele, Thr, Rud, CH5 to CH9 are SA, SB, SC, SD, SE, and CH10 is pot S1	Change the switch on any channel. Rate and expo belong in Betaflight.
Plane	CH1 to CH4 are Ail, Ele, Thr, Rud, CH5 is flaps and CH6 is the second aileron	**2nd aileron** **Flap sw** **V-tail** **Thr cut**
Delta	CH1 is the left wing, CH2 the right wing and CH3 the motor	**L elevon rev** **R elevon rev** **Thr cut**
Heli	CH1, CH2, CH4 are Ail, Ele, Rud, CH3 is throttle, CH5 is flight mode and CH6 is pitch	**Hold sw** **Idle sw** **Mode sw**, three throttle stages and five pitch points

Thr cut on Plane and Delta defaults to SA. The down position cuts the motor. Choose No if you don't want it.

Heli



This is for helicopters with a flybarless controller, which handles the swash itself. The channel layout and defaults come from a Goosky S1. On another brand, check the channels against the controller's manual first.

Hold sw defaults to SA. The down position always cuts the throttle, wherever the stick and other switches are. This row has no No option.

Idle sw defaults to SC. Up is Normal, middle is Idle 1 and down is Idle 2.



If the bottom bar shows SW CLASH in this tool, the wizard won't write to the model. Press **RTN** and separate the switches first.

On a drone, failsafe is set in Betaflight. On Plane, Delta and Heli it is set on the receiver.

Troubleshooting

Problem	Cause and fix
The installer doesn't see the card	The card must come from an EdgeTX radio, with RADIO and SCRIPTS folders. Over USB, choose USB Storage (SD) and wait a moment.
It says Not supported on this radio	Only MT12, Pocket, TX15, TX16S and TX16S MK3 are supported for now.
It says Can't check for updates	There is no internet or the server didn't answer. Go ahead and install: the program uses the latest version it has on this PC.
It says an update failed its check	The downloaded file has no team signature or was damaged on the way, so the program didn't use it. Let TeamSPRC know.
Red text says it can't write the file	The card is locked. Slide the lock switch on the SD adapter away from Lock and press Install again. The files already on the card are unchanged.
No SPRC name in the Tools menu, or it still shows the old name	Open the installer and install again. It removes old .luac files by itself. Remember to eject before removing the card.
The robot doesn't respond at all	The ExpressLRS transmitter and receiver must share a major version, 3.x with 3.x or 4.x with 4.x. Then bind.
The weapon spins when you plug in the battery	Unplug the battery at once and set Weapon ESC to Rev.
The bottom bar shows SW CLASH	Two functions use the same switch. Press RTN and separate them.